Algorithm Sample 5

Dataset-expansion sample

1 Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

function TRANSPORT($\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}, \Delta t$) ▷ Input moments, natural parameters, and gauge parameters for all cells
  $\boldsymbol{\beta} \leftarrow$ NEWTONOPTIMIZATION($\mathbf{M}, \boldsymbol{\beta}, \boldsymbol{\phi}(\mathbf{u}; \mathbf{g})$) ▷ Solve the natural parameters by Alg. **??** in the gauge $\mathbf{g}$
  $\mathbf{F} \leftarrow$ COMPUTEFLUXES($\boldsymbol{\beta}, \boldsymbol{\phi}(\mathbf{u}; \mathbf{g})$) ▷ Compute fluxes of the statistics $\boldsymbol{\phi}(\mathbf{u}; \mathbf{g})$ by (**??**)
  $\mathbf{M}_{\pm 1}, \mathbf{F}_{\pm 1} \leftarrow$ SPATIALGAUGETRANSFORMATION($\mathbf{M}, \mathbf{F}, \mathbf{g}$) ▷ Perform gauge transformation as in (**??**)
  $\mathbf{M} \leftarrow$ FINITEVOLUMESTEP($\mathbf{M}, \mathbf{F}, \mathbf{M}_{\pm 1}, \mathbf{F}_{\pm 1}, \mathbf{g}, \Delta t$) ▷ One step forward in time by (**??**)
  return $\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}$
end function
function COLLISION($\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}, \Delta t$) ▷ Input moments, natural parameters, and gauge parameters for all cells
  $\mathbf{M} \leftarrow$ SOURCETERM($\mathbf{M}, \mathbf{g}, \Delta t$) ▷ Compute source term by operator splitting (**??**)
  return $\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}$
end function
function STEP($\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}$) ▷ Input moments, natural parameters, and gauge parameters for all cells
  $\mathbf{M}, \boldsymbol{\beta}, \mathbf{g} \leftarrow$ COLLISION($\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}, \Delta t/2$) ▷ Compute collision term by operator splitting
  $\mathbf{M}, \boldsymbol{\beta}, \mathbf{g} \leftarrow$ TRANSPORT($\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}, \Delta t$) ▷ Compute Transport term by operator splitting
  $\mathbf{M}, \boldsymbol{\beta}, \mathbf{g} \leftarrow$ COLLISION($\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}, \Delta t/2$) ▷ Compute collision term by operator splitting
  $\mathbf{g}_H \leftarrow$ COMPUTEGAUGEPARAMETERS($\mathbf{M}, \mathbf{g}$) ▷ Compute the Hermite gauge parameters by (**??**)
  $\mathbf{M}_H, \boldsymbol{\beta}_H \leftarrow$ GAUGETRANSFORMATION($\mathbf{M}, \boldsymbol{\beta}, \mathbf{g}_H, \mathbf{g}$) ▷ Transform into the Hermite gauge using (**??**) and (**??**)
  return $\mathbf{M}_H, \boldsymbol{\beta}_H, \mathbf{g}_H$
end function